



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Spring 2022

Paper: Mathematics A-III

Course Code: MATH-201

Roll No.

Time: 3 Hrs.

Marks:

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

(6x5=30)

Q.1. Solve the following:

i) If $|A| \neq 0$, then $|A^n| \neq 0$ for non-zero integer n and $(A^n)^{-1} = (A^{-1})^n$.

ii) Show that a square matrix A satisfies the equation $A^2 + 2A + I = 0$ has $|A| \neq 0$.

iii) Using row operation find A^{-1} if $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$.

iv) Find domain and range $T(x_1, x_2, x_3, x_4) = (x_1, x_2)$.

v) Find area of the triangle with vertices $(3,3), (4,0), (-2,-1)$.

vi) Show $u = (-2, 3, 1, 4)$ & $v = (1, 2, 0, -1)$ are orthogonal vectors in $\mathbb{R}^4$. Also find angle b/w u & v .

Q.2. Solve the following:

(5x6=30)

i) Show that the following system by inverting the coefficient matrix has non trivial solution if

$$x_1 + x_2 + \alpha x_3 = 0; \quad x_1 + \beta x_3 + x_2 = 0; \quad \alpha x_1 + \beta x_2 + x_3 = 0$$

ii) Check either $T(x, y, z) = (x, z, x+y)$ is a matrix transformation also find $R(T)$ and $N(T)$.

iii) A square matrix A is invertible iff $\lambda = 0$ is not an eigen value of A .

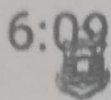
iv) Find the bases for the subspace of $\mathbb{R}^4$ that is spanned by the vectors:

$$(1, 1, -4, -3), (2, 0, 2, -2) \text{ \& } (2, -1, 3, 2)$$

v) What condition must a, b, c and d satisfy so that the following matrices are linearly independent

$$\text{in } M_{22} \quad \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ -2 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} a & b \\ c & d \end{bmatrix}. \quad 241$$

Equation of Spanned Set



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Third Semester – Fall 2021

Paper: Mathematics A-III

Course Code: MATH-201

4G
Roll No.
Time: 3 Hrs. Marks: 60

Q.1. Solve the following:

(6x5=30)

- i. Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x_1, x_2, x_3) = (-x_3, x_1, x_1 + x_3)$. Find $N(T)$. Is T one-to-one?
- ii. Check whether the first quadrant $W := \{(x, y) : x, y \geq 0\}$ makes a subspace of $\mathbb{R}^2$ or not.
- iii. Find the reduced echelon form of the matrix

$$\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$$

- iv. Show that Similar matrices having same eigenvalues.
- v. Show that if $u \in V$ is orthogonal to every $v \in V$, then $u = 0$.
- vi. Give an example of non-similar matrices having same characteristic polynomial.

Q.2. Solve the following:

(5x6=30)

- i. Determine the values of a for which the system of linear equations has no solution, exactly one solution and infinitely many solutions.
 $x + y + 7z = -7$
 $2x + 3y + 17z = -16$
 $x + 2y + (a^2 + 1)z = 3a$

ii.

If possible, find the inverse of the matrix

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$$

iii.

Find Eigen Values and Eigen vectors of

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

iv.

Find solution of the given system by Gauss-Jordan Elimination method.

$$2x_1 - x_2 - x_3 = 4; \quad 3x_1 - 2x_3 + 4x_2 = 11; \quad 3x_1 - 2x_2 + 4x_3 = 11$$

v.

Find an equation of the subspace W of $\mathbb{R}^3$ spanned by $v_1 = (1, -3, 5)$, $v_2 = (-2, 6, -10)$.



UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program Third Semester – 2020

Roll No.

Paper: Mathematics A-III

(CLASH)

Time: 2 Hrs. 30 Min. Marks: 50

Course Code: MATH-201/MTH-21308 Part - II

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Solve the following:

(5x4=20)

(i)	Determine whether the vectors are linearly independent or not? $v_1 = (1, -2, 3), v_2 = (5, 6, -1), v_3 = (3, 2, 1)$	(4)
(ii)	If $(A^T - 2I)^{-1} = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$ then find A .	(4)
(iii)	Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x_1, x_2, x_3) = (-x_1, x_2, x_1 + x_3)$. Find $N(T)$. Is T one-to-one? <i>mobile may be</i>	(4)
(iv)	Check whether W is a subspace of V or not. $V = \{f: f: \mathbb{R} \rightarrow \mathbb{R}\}, W = \{f \in V: f(1) = 0\}$. <i>5/10 2d9 (iii)</i>	(4)
(v)	Check whether the functions $f(t) = \tan t, g(t) = \sin t, h(t) = \cos t$ from $\mathbb{R}$ to $\mathbb{R}$ are linearly independent. <i>وَقُلْ رَبِّ زِدْنِي عِلْمًا</i>	(4)

Solve the following:

(5x6=30)

Q.3	Solve the system of linear equations $2x + y + z = 1, 3x + y - 5z = 8, 4x - y + z = 5$	(6)
Q.4	If possible, find the inverse of the matrix $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$	(6)
Q.5	Show that $\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} = x^2 y^2$	(6)
Q.6	(b) Inside $\mathbb{R}^3$, decide which of the following set(s) is (are) linearly independent $\{v_1, v_2\}, \{v_2, v_3, v_4\}, \{v_3, v_4\}$, where $v_1 = (0, 0, 0), v_2 = (1, 0, 1), v_3 = (1, -1, 0), v_4 = (1, 0, -1)$.	(6)
Q.7	Find the basis of Null space associated with the given matrix. $\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 0 \end{bmatrix}$	(6)

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$n = 4$$

$$m = n = 3$$

$$T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$$

$$T(x) = Ax$$

$$u \in \mathbb{R}^4 = \{x_1, x_2, x_3, x_4\}$$



UNIVERSITY OF THE PUNJAB

Third Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Mathematics A-III

Course Code: MATH-201/MTH-21309 Part – II

MAX. TIME: 2 Hrs. 30 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Questions with Short Answers.

(5x4=20)

(i)	Decide yes or no about the following statement, if no justify by counter example. If A is 2×2 matrix such that $A^2 = 0$ Then A must be a null matrix-	(4)
(ii)	If possible express $\vec{v} = (2, -5, 3)$ in R^3 as a linear combination of the vectors $\vec{u}_1 = (1, -3, 2), \vec{u}_2 = (2, -4, 1), \vec{u}_3 = (1, 5, 7)$.	(4)
(iii)	Check whether W is a subspace of V or not. $V = \{f : f : R \rightarrow R\}, W = \{f \in V : f(1) = 0\}$.	(4)
(iv)	Show that $\det \begin{bmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{bmatrix} = (a-1)^3(a+3)$	(4)
(v)	If A and B are invertible matrices of same order and AB is invertible then show that $(AB)^{-1} = B^{-1}A^{-1}$.	(4)

P.T.O.

LONG QUESTIONS

(5x6=30)

Solve the following system of linear equations by using an appropriate method.

Q.3

$$\begin{aligned}x_1 - 2x_2 + 3x_3 &= 3 \\2x_1 + x_2 - x_3 &= 9 \\-3x_1 - 4x_2 + 5x_3 &= -15.\end{aligned}$$

Q.4

Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x_1, x_2, x_3) = (-x_1, x_1, x_1 + x_2)$. Find $N(T)$. Is T one-to-one?

Q.5

If $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ determine the value of $A^3 - 4A - 5I$. *book 127*

Q.6

Find the real orthogonal matrix P for which $P^{-1}AP$ is orthogonal where

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

Q.7

Express matrix M as a linear combination of the matrices A , B and C

where

$$M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}.$$



UNIVERSITY OF THE PUNJAB

Third Semester 2018
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Mathematics A-III

Course Code: MATH-201/MTM-21309

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE SECTION - II

Q. 2	SHORT QUESTIONS	
(i)	For the given matrix find the basis of Column space. $\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix}$	(4)
(ii)	If A is a nonsingular matrix whose inverse is $\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, find A.	(4)
(iii)	Find the reduced echelon form of the matrix $\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$	(4)
(iv)	Prove that $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$	(4)
(v)	Check whether W is a subspace of V or not? $V = \mathbb{R}^3$, $W = \{(a, b, c) \in V : a^2 + b^2 + c^2 \leq 1\}$.	(4)

SECTION - III

LONG QUESTIONS		
Q.3	If possible, find the inverse of the matrix $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$	(6)
Q.4	Determine whether the vectors are linearly independent or not? $v_1 = (1, -2, 3)$, $v_2 = (5, 6, -1)$, $v_3 = (3, 2, 1)$.	(6)
Q.5	Determine whether or not the set of vectors $\{(1, 2, -1), (0, 3, 1), (1, -5, 3)\}$ is a basis for $\mathbb{R}^3$	(6)
Q.6	Find the real orthogonal matrix P for which $P^{-1}AP$ is orthogonal where $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$	(6)
Q.7	Determine the values of a for which the system of linear equations has no solution, exactly one solution and infinitely many solutions. $x + y + 7z = -7$ $2x + 3y + 17z = -16$ $x + 2y + (a^2 + 1)z = 3a$.	(6)



Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q2. Short Questions (2x10=20)

- i) Let A and B be 3×3 matrices with $\det(A) = 2$ and $\det(B) = 4$. Find the values of $\det(AB)$ and $\det(A^{-1}B)$.
- ii) Consider a linear system whose augmented matrix is of the form

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ -1 & 4 & 3 & 2 \\ 2 & -2 & a & 3 \end{array} \right] \text{ For what value(s) of } a \text{ will the}$$

system have a unique solution?

- iii) Define Undetermined system.
- iv) Define determinant of a matrix.
- v) Check whether the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined as $T(x, y) = (-y, x)$ is linear or not.
- vi) Check whether $W = \{(x, 0, z) : x, y, z \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$?
- vii) Find the eigen values of linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined as $T(x, y) = (3x + 3y, x + 5y)$.
- viii) Without expansion, show that
- $$\begin{vmatrix} a+b & c & 1 \\ b+c & a & 1 \\ c+a & b & 1 \end{vmatrix} = 0$$
- ix) Show that if A is a symmetric nonsingular matrix, then A^{-1} is also symmetric.
- x) Let $A = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$
Compute A^T and A^{-1} . What will A^{-1} turn out to be?

Subjective Questions (5x6=30)

Q3. Determine the solution of the following system of linear equations

$$\begin{aligned} x - y + z &= 1 \\ x - 2y - 3z &= 6 \\ 2x - y - z &= 6 \end{aligned}$$

Q4. Find equations defining the subspace W of $\mathbb{R}^3$ spanned by the vector $(2, 3, 4)$.

Q5. Check whether the following transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is one-one or not
 $T(x, y, z) = (x - y, z)$.

Q6. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by
 $T(a, b, c) = (a + 2b - c, b + c, a + b - 2c)$. Find basis and dimension of $R(T)$ and $N(T)$.

Q7. Prove that a one-one linear transformation preserves basis and dimension.

Q8. If A is a matrix over $\mathbb{R}$ and $AA^T = 0$, then show that $A = 0$.

UNIVERSITY OF THE PUNJAB



Third Semester 2015
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Mathematics A-III
Course Code: MATH-101

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Short Questions

(2x10=20)

Q2.

- Define a "Nilpotent matrix"
- Define a "Hermitian matrix"
- Let $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$, $|A| \neq 0$, $A \neq I$, then find a , b and c if A is involutory.
- If A is a 3×3 matrix with $\det A = 2$, then find $\det A^3$.
- Prove that $\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$
- Show that the set $\{(1,3), (2,5)\}$ is linearly independent but the set $\{(1,3), (2,6)\}$ is linearly dependent in $\mathbb{R}^2$.
- Check whether $W = \{(x,y,z) : x+y+z=0\}$ is a subspace of $\mathbb{R}^3$.
- Find the dimension of subspace $\mathbb{R}^4$, spanned by $\{(-1, -1, 5, 0), (0, 0, 0, 1)\}$.
- Find the eigen values for the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, such that $T(x,y) = (3x+3y, x+5y)$
- Show that A and A^t have same eigen values, where A is a square matrix.

Subjective Questions

(6x5=30)

- Q3. Use Gauss-Jordan method to reduce the given system to reduced echelon form and determine solution.

$$6x - 6y + 6z = 6$$

$$2x - 4y - 6z = 12$$

$$10x - 5y - 5z = 30$$

- Q4. If A and B are 3×3 matrices such that $\det(A^2 B^2) = 108$ and $\det(A^2 B^3) = 72$, find $\det(2A)$ and $\det(B^4)$.
- Q5. Determine 'k' so that the vectors $(1, -1, k-1)$, $(2, k, -4)$, $(0, 2+k, -8)$ in $\mathbb{R}^3$ are linearly independent.
- Q6. Let U and W be 2-dimensional subspaces of $\mathbb{R}^3$, show that $U \cap W = \{0\}$.
- Q7. Show that the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x,y) = (x+y, x-y, x+2y)$ is one-one.

- Q8. Find an orthogonal matrix whose first row is $(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.